

Indirect Tax Classification & Reconciliation Engine - Executive Summary

A rules-based engine that independently derives the correct indirect-tax treatment for an AP invoice line and flags it when the vendor-master tax code disagrees - the exact mechanic behind a tax-ops exception queue.

Headline result

- **20 rules sourced across 11 jurisdictions** - India GST (post-Sept-2025 4-slab reform) and TDS, US state Sales & Use Tax (CA/NY/IL/TX), EU VAT (DE/FR/LU/HU/IE/NL) - each with a citation and an as_of_date.
- **60 synthetic invoice lines, engine-classified with a 33.3% exception rate** - exactly matching the 1-in-3 error-injection rate built into the test set, confirming the classification logic is internally consistent (see Recommendation Memo, "What broke").
- **9 of 20 rules are flagged** `stable_fact_verify_before_filing` rather than presented with false confidence - the engine itself demotes any line relying on one of these to OK - VERIFY RATE instead of a clean pass.

Resume-ready line

"Built a rules-based indirect-tax classification engine spanning GST, TDS (India), Sales & Use Tax (US), and VAT (EU) across 11 jurisdictions and 20 sourced rules; tested against 60 synthetic invoice lines with a 33.3% exception-detection rate, independently verified to match the injected error rate exactly."

What this is not

Not a real tax filing tool, not trained on or tested against any real vendor data (none exists for this project), and not a claim of SAP/Vertex/OneSource software experience. It demonstrates the underlying reasoning a tax-ops analyst applies - jurisdiction + transaction-type -> correct treatment -> flag the mismatch - not production tax software.